

Course Overview

As Artificial Intelligence (AI) becomes integrated into high-risk domains like healthcare, finance, and criminal justice, it is critical that those responsible for building these systems think outside the black box and develop systems that are not only accurate, but also transparent and trustworthy. This course is a comprehensive, hands-on guide to Explainable Machine Learning (XAI), empowering you to develop AI solutions that are aligned with responsible AI principles.

Through discussions, case studies, programming labs, and real-world examples, you will gain the following skills:

1. Implement local explainable techniques like LIME, SHAP, and ICE plots using Python.
2. Implement global explainable techniques such as Partial Dependence Plots (PDP) and Accumulated Local Effects (ALE) plots in Python.
3. Apply example-based explanation techniques to explain machine learning models using Python.
4. Visualize and explain neural network models using SOTA techniques in Python.
5. Critically evaluate interpretable attention and saliency methods for transformer model explanations.
6. Explore emerging approaches to explainability for large language models (LLMs) and generative computer vision models.

This course is ideal for data scientists or machine learning engineers who have a firm grasp of machine learning but have had little exposure to XAI concepts. By mastering XAI approaches, you'll be equipped to create AI solutions that are not only powerful but also interpretable, ethical, and trustworthy, solving critical challenges in domains like healthcare, finance, and criminal justice.

To succeed in this course, you should have an intermediate understanding of machine learning concepts like supervised learning and neural networks.

Note on Programming Assignments

All programming assignments in this course are ungraded, but they are an essential part of your learning experience, providing helpful hands-on practice. The majority of the labs are housed in Coursera's integrated lab environment, ensuring a seamless and interactive experience. For the handful of labs that require Google Colab, detailed instructions are provided to guide you on how to access and utilize these resources effectively. Engaging with these labs will deepen your understanding and enhance your skills in implementing explainable machine learning techniques. All labs in this course are accessible free of cost to learners.

Coursera

If you have any Coursera-Specific questions about navigation or assignment types, the best resource would be the Coursera Learner Help Center. You can click the small question mark icon in the bottom right of your screen at any time to access help.

Certificate vs. Audit Experience

The required assignments for this course to earn the certificate are graded quiz activities.

You will only be able to submit graded quizzes if you are enrolled to earn the certificate, rather than auditing the course.

As a learner enrolled for a certificate, you will be able to access:

- The course quizzes, which means you will be able to submit the quizzes and receive automatic feedback on your answers.
- All other course content (videos, readings, labs, and discussions).

As an auditing learner, you will only be able to access the videos, readings, labs, and discussions. You can view the graded quizzes, but you will not be able to submit your answers and receive feedback.

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